

10)

AIM:- Creating graphical reports using ,ggplot2 (R).

INPUT:-

```
library(ggplot2)
```

```
library(dplyr)
```

```
setwd("C:/Users/ASUS/Downloads")
```

```
shades <- read.csv("shades (1).csv")
```

```
str(shades)
```

```
head(shades)
```

```
ggplot(shades, aes(x = brand)) +
```

```
  geom_bar() +
```

```
  labs(
```

```
    title = "Number of Shades by Brand",
```

```
    x = "Brand",
```

```
    y = "Number of Shades"
```

```
  ) +
```

```
  theme_minimal() +
```

```
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
ggplot(shades, aes(x = product)) +
```

```
  geom_bar() +
```

```
  labs(
```

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```
title = "Number of Shades by Product",  
x = "Product",  
y = "Number of Shades"  
) +  
theme_minimal() +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))  
ggplot(shades, aes(x = brand, y = L)) +  
geom_boxplot() +  
labs(  
  title = "Lightness (L) Distribution by Brand",  
  x = "Brand",  
  y = "Lightness (L)"  
) +  
theme_minimal() +  
theme(axis.text.x = element_text(angle = 45, hjust = 1))  
ggplot(shades, aes(x = H, y = L)) +  
geom_point(alpha = 0.7) +  
labs(  
  title = "Hue vs Lightness",  
  x = "Hue (H)",  
  y = "Lightness (L)"  
) +  
theme_minimal()  
ggplot(shades, aes(x = factor(group))) +  
geom_bar() +  
labs()
```

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```
title = "Number of Shades by Group",  
  
x = "Shade Group",  
  
y = "Number of Shades"  
  
) +  
  
theme_minimal()  
  
ggplot(shades, aes(x = S)) +  
  
geom_histogram(bins = 20) +  
  
labs(  
  
  title = "Distribution of Saturation (S)",  
  
  x = "Saturation",  
  
  y = "Count"  
  
) +  
  
theme_minimal()
```

OUTPUT:-

```
Console Terminal Background Jobs  
R 4.5.2 · C:/Users/ASUS/Downloads/ ↗  
  
> library(ggplot2)  
> library(dplyr)  
>  
> setwd("C:/Users/ASUS/Downloads")  
>  
> shades <- read.csv("shades (1).csv")  
>  
> str(shades)  
'data.frame': 625 obs. of 10 variables:  
 $ brand      : chr  "Maybelline" "Maybelline" "Maybelline" "Maybelline" ...  
 $ brand_short: chr  "mb" "mb" "mb" "mb" ...  
 $ product    : chr  "Fit Me" "Fit Me" "Fit Me" "Fit Me" ...  
 $ product_short: chr  "fmf" "fmf" "fmf" "fmf" ...  
 $ hex        : chr  "f3cfb3" "ffe3c2" "ffe0cd" "ffd3be" ...  
 $ H          : int   26 32 23 19 18 20 28 24 26 20 ...  
 $ S          : num   0.26 0.24 0.2 0.25 0.3 0.29 0.31 0.33 0.38 0.38 ...  
 $ V          : num   0.95 1 1 1 0.74 0.92 0.98 0.89 0.89 0.7 ...  
 $ L          : int   86 92 91 88 65 80 87 77 77 60 ...  
 $ group      : int   2 2 2 2 2 2 2 2 2 2 ...  
> head(shades)  
  brand brand_short product product_short hex H S V L group  
1 Maybelline      mb Fit Me      fmf f3cfb3 26 0.26 0.95 86 2  
2 Maybelline      mb Fit Me      fmf ffe3c2 32 0.24 1.00 92 2  
3 Maybelline      mb Fit Me      fmf ffe0cd 23 0.20 1.00 91 2  
4 Maybelline      mb Fit Me      fmf ffd3be 19 0.25 1.00 88 2  
5 Maybelline      mb Fit Me      fmf bd9584 18 0.30 0.74 65 2  
6 Maybelline      mb Fit Me      fmf eabda6 20 0.29 0.92 80 2  
>  
> ggplot(shades, aes(x = brand)) +  
+   geom_bar() +  
+   labs(  
+     title = "Number of Shades by Brand",  
+     x = "Brand",  
+     y = "Number of Shades"  
+   ) +  
+   theme_minimal() +  
+   theme(axis.text.x = element_text(angle = 45, hjust = 1))  
> ggplot(shades, aes(x = product)) +  
+   geom_bar() +  
+   labs(  
+     title = "Number of Shades by Product",  
+     x = "Product",  
+     y = "Number of Shades"  
+   ) +  
+   theme_minimal()
```

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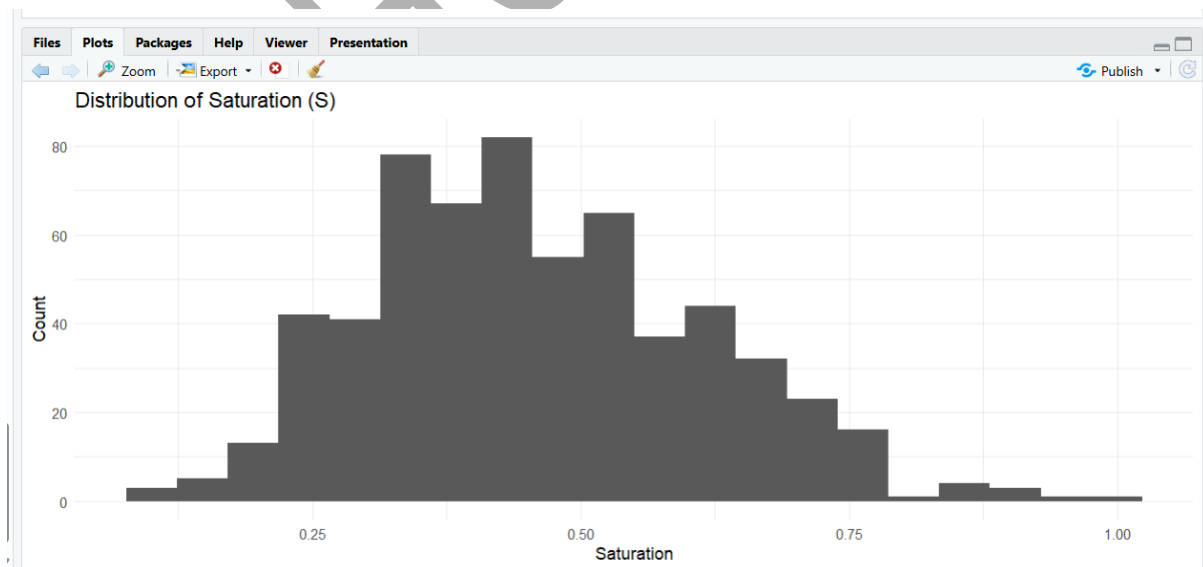
```
Console Terminal Background Jobs
R - R 4.5.2 · C:/Users/ASUS/Downloads/
+ labs(
+   title = "Number of Shades by Product",
+   x = "Product",
+   y = "Number of Shades"
+ ) +
+ theme_minimal() +
+ theme(axis.text.x = element_text(angle = 45, hjust = 1))
> ggplot(shades, aes(x = brand, y = L)) +
+ geom_boxplot() +
+ labs(
+   title = "Lightness (L) Distribution by Brand",
+   x = "Brand",
+   y = "Lightness (L)"
+ ) +
+ theme_minimal() +
+ theme(axis.text.x = element_text(angle = 45, hjust = 1))
> ggplot(shades, aes(x = H, y = L)) +
+ geom_point(alpha = 0.7) +
+ labs(
+   title = "Hue vs Lightness",
+   x = "Hue (H)",
+   y = "Lightness (L)"
+ ) +
+ theme_minimal()

Warning message:
Removed 12 rows containing missing values or values outside the scale range ('geom_point()').

> ggplot(shades, aes(x = factor(group))) +
+ geom_bar() +
+ labs(
+   title = "Number of Shades by Group",
+   x = "Shade Group",
+   y = "Number of Shades"
+ ) +
+ theme_minimal()
> ggplot(shades, aes(x = S)) +
+ geom_histogram(bins = 20) +
+ labs(
+   title = "Distribution of Saturation (S)",
+   x = "Saturation"
+ ) +
+ theme_minimal()

Warning message:
Removed 12 rows containing non-finite outside the scale range ('stat_bin()').

> |
```



11)

AIM:- Generating histograms and box plots using ggplot2 (R).

INPUT:-

```
library(ggplot2)
```

```
iris_data <- read.csv("C:/Users/ASUS/Downloads/IRIS (1).csv")
```

```
head(iris_data)
```

```
str(iris_data)
```

```
ggplot(iris_data, aes(x = sepal_length)) +
```

```
  geom_histogram(binwidth = 0.3) +
```

```
  labs(
```

```
    title = "Histogram of Sepal Length",
```

```
    x = "Sepal Length",
```

```
    y = "Frequency"
```

```
  ) +
```

```
  theme_minimal()
```

```
ggplot(iris_data, aes(x = sepal_length)) +
```

```
  geom_histogram(binwidth = 0.3) +
```

```
  facet_wrap(~ species) +
```

```
  labs(
```

```
    title = "Histogram of Sepal Length by Species",
```

```
    x = "Sepal Length",
```

```
    y = "Frequency"
```

```
  ) +
```

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```
theme_minimal()

ggplot(iris_data, aes(y = sepal_length)) +

  geom_boxplot() +

  labs(

    title = "Box Plot of Sepal Length",

    y = "Sepal Length"

  ) +

  theme_minimal()

ggplot(iris_data, aes(x = species, y = sepal_length)) +

  geom_boxplot() +

  labs(

    title = "Box Plot of Sepal Length by Species",

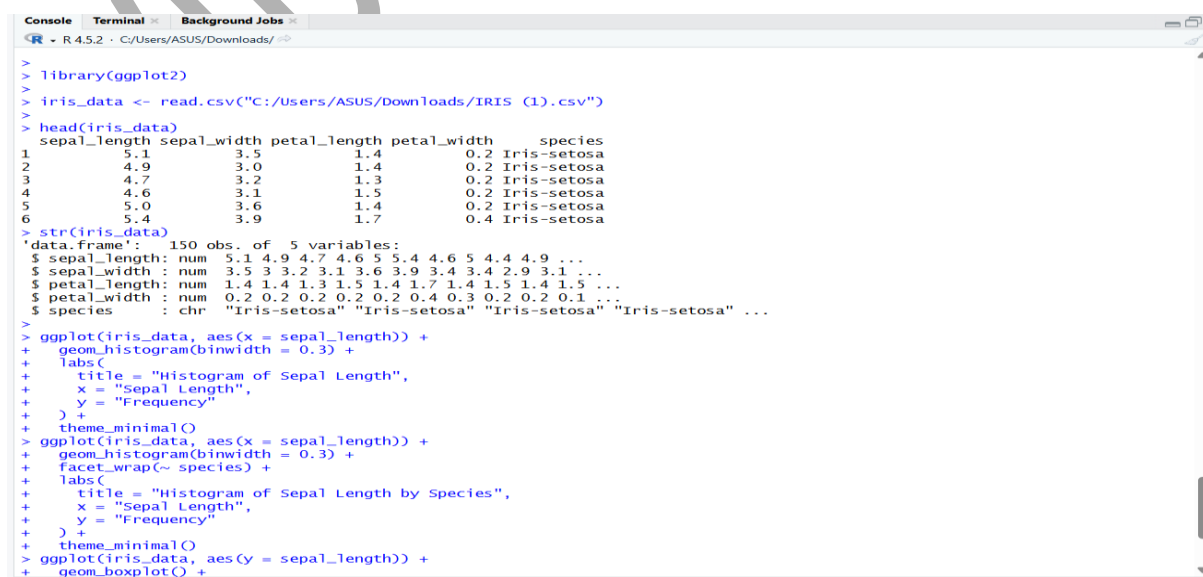
    x = "Species",

    y = "Sepal Length"

  ) +

  theme_minimal()
```

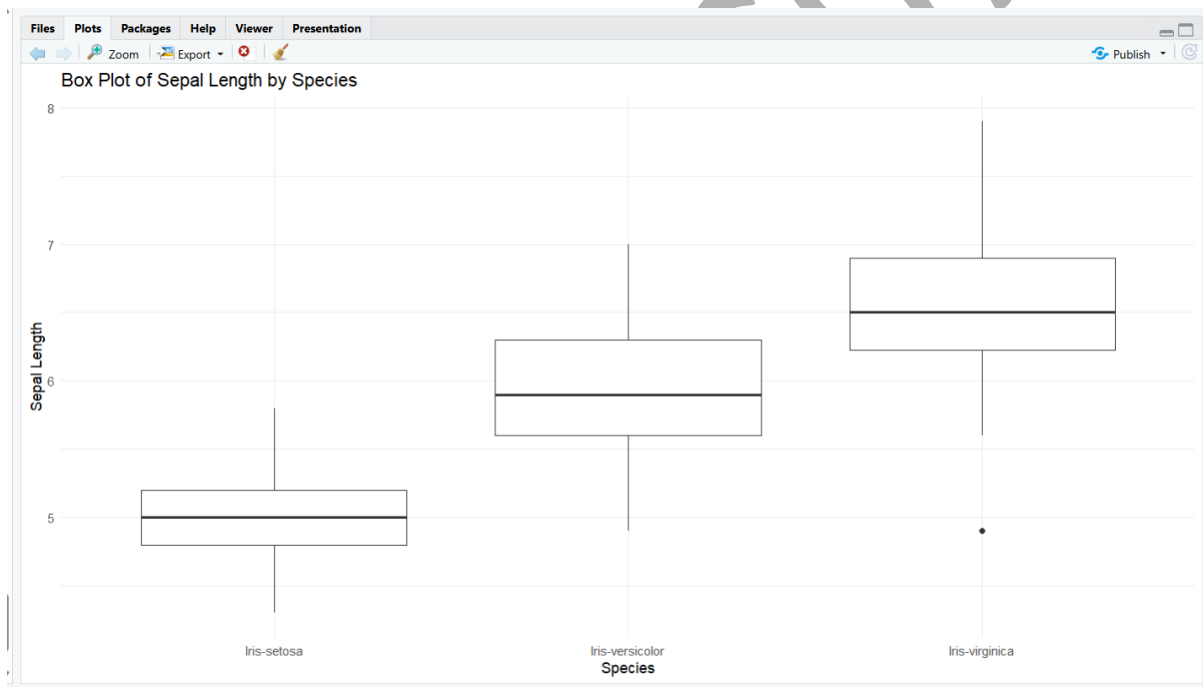
OUTPUT:-



```
Console Terminal Background Jobs
R 4.5.2 - C:/Users/ASUS/Downloads/
> library(ggplot2)
> iris_data <- read.csv("C:/Users/ASUS/Downloads/IRIS (1).csv")
> head(iris_data)
  sepal_length sepal_width petal_length petal_width  species
1         5.1         3.5         1.4         0.2 Iris-setosa
2         4.9         3.0         1.4         0.2 Iris-setosa
3         4.7         3.2         1.3         0.2 Iris-setosa
4         4.6         3.1         1.5         0.2 Iris-setosa
5         5.0         3.6         1.4         0.2 Iris-setosa
6         5.4         3.9         1.7         0.4 Iris-setosa
> str(iris_data)
'data.frame':   150 obs. of  5 variables:
 $ sepal_length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ sepal_width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ petal_length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ petal_width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ species     : chr  "Iris-setosa" "Iris-setosa" "Iris-setosa" "Iris-setosa" ...
> ggplot(iris_data, aes(x = sepal_length)) +
+   geom_histogram(binwidth = 0.3) +
+   labs(
+     title = "Histogram of Sepal Length",
+     x = "Sepal Length",
+     y = "Frequency"
+   ) +
+   theme_minimal()
> ggplot(iris_data, aes(x = sepal_length)) +
+   geom_histogram(binwidth = 0.3) +
+   facet_wrap(~ species) +
+   labs(
+     title = "Histogram of Sepal Length by Species",
+     x = "Sepal Length",
+     y = "Frequency"
+   ) +
+   theme_minimal()
> ggplot(iris_data, aes(y = sepal_length)) +
+   geom_boxplot() +
```

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```
+ title = "Histogram of Sepal Length",  
+ x = "Sepal Length",  
+ y = "Frequency"  
+ ) +  
+ theme_minimal()  
> ggplot(iris_data, aes(x = sepal_length)) +  
+ geom_histogram(binwidth = 0.3) +  
+ facet_wrap(~ species) +  
+ labs(  
+ title = "Histogram of Sepal Length by Species",  
+ x = "Sepal Length",  
+ y = "Frequency"  
+ ) +  
+ theme_minimal()  
> ggplot(iris_data, aes(y = sepal_length)) +  
+ geom_boxplot() +  
+ labs(  
+ title = "Box Plot of Sepal Length",  
+ y = "Sepal Length"  
+ ) +  
+ theme_minimal()  
> ggplot(iris_data, aes(x = species, y = sepal_length)) +  
+ geom_boxplot() +  
+ labs(  
+ title = "Box Plot of Sepal Length by Species",  
+ x = "Species",  
+ y = "Sepal Length"  
+ ) +  
+ theme_minimal()  
>
```



12

AIM:- Generating correlation matrices using cor() (R).

INPUT:-

```
shades <- read.csv("shades (1).csv")
```

```
head(shades)
```

```
str(shades)
```

```
numeric_data <- shades[sapply(shades, is.numeric)]
```

```
head(numeric_data)
```

```
correlation_matrix <- cor(numeric_data, use = "complete.obs")
```

```
print(correlation_matrix)
```

```
rounded_correlation <- round(correlation_matrix, 2)
```

```
print(rounded_correlation)
```

OUTPUT:-

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```

Console Terminal Background Jobs
R - R 4.5.2 - C:/Users/ASUS/Downloads/
> shades <- read.csv("shades (1).csv")
>
> head(shades)
  brand brand_short product product_short hex H S V L group
1 Maybelline      mb Fit Me          fmf f3cfb3 26 0.26 0.95 86 2
2 Maybelline      mb Fit Me          fmf ffe3c2 32 0.24 1.00 92 2
3 Maybelline      mb Fit Me          fmf ffe0cd 23 0.20 1.00 91 2
4 Maybelline      mb Fit Me          fmf ffd3be 19 0.25 1.00 88 2
5 Maybelline      mb Fit Me          fmf bd9584 18 0.30 0.74 65 2
6 Maybelline      mb Fit Me          fmf eabda6 20 0.29 0.92 80 2
> str(shades)
'data.frame':   625 obs. of  10 variables:
 $ brand      : chr  "Maybelline" "Maybelline" "Maybelline" "Maybelline" ...
 $ brand_short: chr  "mb" "mb" "mb" "mb" ...
 $ product    : chr  "Fit Me" "Fit Me" "Fit Me" "Fit Me" ...
 $ product_short: chr  "fmf" "fmf" "fmf" "fmf" ...
 $ hex        : chr  "f3cfb3" "ffe3c2" "ffe0cd" "ffd3be" ...
 $ H          : int   26 32 23 19 18 20 28 24 26 20 ...
 $ S          : num  0.26 0.24 0.2 0.25 0.3 0.29 0.31 0.33 0.38 ...
 $ V          : num  0.95 1 1 1 0.74 0.92 0.98 0.89 0.89 0.7 ...
 $ L          : int   86 92 91 88 65 80 87 77 60 ...
 $ group      : int   2 2 2 2 2 2 2 2 2 2 ...
> numeric_data <- shades[is.numeric(shades)]
>
> head(numeric_data)
  H S V L group
1 26 0.26 0.95 86 2
2 32 0.24 1.00 92 2
3 23 0.20 1.00 91 2
4 19 0.25 1.00 88 2
5 18 0.30 0.74 65 2
6 20 0.29 0.92 80 2
> correlation_matrix <- cor(numeric_data, use = "complete.obs")
>
> print(correlation_matrix)
      H          S          V          L          group
H  1.0000000 -0.16643623 0.4098312 0.4514155 0.11856059
S -0.1664362 1.00000000 -0.7077968 -0.8106189 -0.04826675
V  0.4098312 -0.70779678 1.0000000 0.9806898 0.16553491
L  0.4514155 -0.81061885 0.9806898 1.0000000 0.14590430
group 0.1185606 -0.04826675 0.1655349 0.1459043 1.00000000
>
> print(correlation_matrix)
      H          S          V          L          group
H  1.0000000 -0.16643623 0.4098312 0.4514155 0.11856059
S -0.1664362 1.00000000 -0.7077968 -0.8106189 -0.04826675
V  0.4098312 -0.70779678 1.0000000 0.9806898 0.16553491
L  0.4514155 -0.81061885 0.9806898 1.0000000 0.14590430
group 0.1185606 -0.04826675 0.1655349 0.1459043 1.00000000
> rounded_correlation <- round(correlation_matrix, 2)
>
> print(rounded_correlation)
      H          S          V          L          group
H  1.00 -0.17 0.41 0.45 0.12
S -0.17 1.00 -0.71 -0.81 -0.05
V  0.41 -0.71 1.00 0.98 0.17
L  0.45 -0.81 0.98 1.00 0.15
group 0.12 -0.05 0.17 0.15 1.00
>

```